ROYAL CANADIAN NAVY QUALIFICATION STANDARD AND PLAN (QSP)

Maritime Technician (MAR TECH) (00379-03)

USQ-PO2 Electrical Supervisor (Elec Sup)

MITE Course IDs: 122004
NQual: ALUY

RECORD OF AMENDMENTS

1. This QSP is developed under the authority of the Royal Canadian Navy (RCN) through the Training Authority (TA), Naval Personnel and Training Group (NPTG) and the Commanding Officer of the Naval Training Development Centre (NTDC). The approval or revision date should appear in the footer of every page as an additional means of verification. Details on the change(s) must be amplified in Annex J "Course History".

Revision	Approval Date	Authority
1.0 (QSPWB Apr 18; review Apr 22)		NTDC(P) CO
	10 May 2024	CNPTG

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MARPAC RDIMS #823508
Chap, 1, approved 10 05 2024
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FOREWORD

1. This training document is developed under the authority of the Commander Naval Personnel and Training Group (CNPTG) and in accordance with the following references:

   a. A-P9 – 050 Series, Canadian Forces Manual of Individual Training and Education (CFITES);

   b. A-PD-055-002/PP-004 Non-Commissioned Member General Specification 9 Feb 2011;

   c. A-PD-055-002/PP-001 5555-00379-MAR-TECH Occupational Specification dated 15 Dec 16;

   d. NAVORD 4500-0 “Structured Delegation”; and

   e. NPTORD 4500-22 “Qualification Standard and Plan Production and Management”.

2. This QSP consists of specific Performance Objectives (POs), Enabling Objectives (EOs), and Lesson Specifications in Chapter 3. A trainee will achieve the qualification when the standards listed in the POs have been met.

3. This QSP is effective upon approval and supersedes previous versions.

4. Suggestions for amendments to this QSP may be initiated by anyone and shall be forwarded in writing to the CCA (NTDC(P) MSE), who will forward to NPTG for approval, as appropriate.

5. The designated TEs for this training are indicated below:

TE	Phase / Module	MITE Course ID	NQual
NFS(P)	All	122004	ALUY
NFS(A)	All	122004	ALUY

CHAPTER 1 TRAINING MANAGEMENT DETAILS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY
COMMANDER NAVAL PERSONNEL AND TRAINING GROUP (CNPTG)

AIM

6. The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training for the Electrical Supervisor (Elec Sup). This QSP is designed to assist training establishments in developing, conducting, evaluating, and validating the training.

COURSE DURATION

7. Course duration is calculated in accordance with NPTORD 4500-22 Annex B. Detailed duration with resource allocation is found in Annex G of this QSP.

Allocation of ATT	**Minutes**
Sum of all EOs/POs	1500
Enabling Checks (ECs)	430
Performance Checks (PCs)	460
Actual Training Time (ATT) Sub-total	**2390**
Allocation of BAT, TAT, TTT, PT	
Base Administrative Time (BAT)	60
Training Administrative Time (TAT)	60
Training Travel Time (TTT)	0
Physical Fitness (PT)	0
Sub-total	120
Course Duration Calculation (ATT+BAT+TTT+TAT+PT)	2510
Total mins/400 = # days (rounded up)	
Training Days	**7**

RESOURCES

8. The following table details key personnel required to conduct the training:

Required Course Staffing				
Position	Rank	Qualifications Required	Qty	PO / EO
Sr Instructor	PO1	Post Electrical Manager at sea	1	All
PCOT operator	MS-PO2	NFS PCOT Operations Staff	1	All
Guest Speaker	Lt(N)-LCdr	MSEO Post HOD tour	1	001.01
Guest Speaker	LCdr - Cdr	NWO - Command Qualified	1	002.01

9. The following table details the critical training resources to conduct the training. It should be read in conjunction with the recommended course timetable in Annex G of this QSP. Amplifications for resource requirements are found at Annex H.

Critical Training Resources	PO/EO
DWAN Laptop (each trainee)	All
DWAN drops	All
Classroom with projector	All
PCOT (see annex H for details)	All

COURSE CAPACITY

10. Planning for the conduct of training is based on the guidelines in CFITES Vol 5 Annex D (Instructor to Learner Ratios). Capacity is limited by stations available in the PCOT.

    a. Maximum – 12;

    b. Minimum – 4; and

    c. Optimum – 12.

MITE

11. Loading. The authority to load members onto this course lies with the NFS. NPTG is to be included as an INFO addressee on all loading messages.

12. Scheduling. It is to be scheduled in MITE by the NFS conducting the training and NPTG through the Quantity Control section. The course length in MITE is to be based on the course duration, planned annual leave, training breaks, weekends and training system holidays. Course sessions for Naval Reserve Force may be scheduled during training system holidays.

13. Prerequisites. The loading authority must ensure that trainees have the following prerequisites before being selected for this course:

    a. Minimum rank PO2;

    b.

Qualification	MITE Code	NQual
USQ-MS SENIOR ELECTRICAL SPECIALIST	122021	ALUZ
RQ-PO2 MAR TECH	121982	ALUW

COURSE REPORTING

14. Upon passing this course, trainees shall receive a Course Report (CF 377) and Certificate of Military Achievement (CF 289) completed IAW DAOD 5031-9 “Course Reporting and Certificates” and NPTORD 4500-51 “Course Reports and Certification”.

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

15. PLARs shall be conducted in accordance with NAVORD 4500-2 “RCN Prior Learning Assessment and Recognition”, DAOD 5031-1 “Canadian Forces Military Equivalencies Program”, and CFITES Vol 12 “Canadian Forces Military Equivalencies Programme: Prior Learning Assessment”.

SPECIAL INSTRUCTIONS

16. Learning Objectives throughout this QSP have been designated as high priority for training as each PO and EO incorporates elements which could have a severe impact to equipment or personnel safety.

CHAPTER 2 ASSESSMENT OF TRAINEE

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER OF THE TRAINING DEVELOPMENT CENTRE

GENERAL

1. This is a pass/fail course. Trainees must successfully complete all POs to pass this course. Performance Checks (PCs), and Enabling Checks (ECs), are based on the Performance Objectives (POs) and Enabling Objectives (EOs) in Chapter 3 of this QSP. All PCs must be successfully completed to achieve the qualification.

2. Trainees will be developed and assessed using a combination of formative and summative assessment events. At a minimum, these will involve PCs, ECs, end of lesson confirmation, homework, assignments, and/or quizzes. See Annex K for details.

ENABLING AND PERFORMANCE CHECKS

3. Enabling Checks (ECs). ECs are formative assessments that help the trainee and the instructors recognize progress or gaps in learning so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. ECs also reinforce learning to improve retention of course content. Formative assessments are designed to provide feedback to both instructional staff and trainees. Each EO must be formatively assessed prior to summative assessments.

4. In the event of an EC failure, the trainee will be counselled and given remedial training and/or additional homework, as appropriate. In the event of a consecutive EC failure, the trainee will be counselled IAW CFITES vol 6 with the unsatisfactory progress documented in their Trainee File.

5. Performance Checks (PCs). PCs are summative tests used to determine if the trainee has met the standard under the conditions (where possible) of the PO and hence can do the job. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

6. PCs will be conducted after formative assessment has occurred and the trainee has had the opportunity to develop skills and apply knowledge related to the PO.

7. All PC failures and supplemental failures are to be handled IAW NPTORD 4500-54 “Training Review Board Process”. Supplemental PCs will normally be scheduled not less than 24 hours post counselling, depending on assessment resource availability.

RESPONSIBILITY FOR TESTING

8. Assessment of trainees is to be done IAW CFITES Vol. 7 “Evaluation of Leaners”.

9. NFS staff provides quality control by administering and scoring all ECs and PCs IAW the standards laid out in this QSP and the assessments developed by NTDC (P).

PROGRESS MONITORING

10. Trainee progress will be monitored using formative assessment tools to help the trainee and instructional staff recognize progress or lapses in learning so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. Formative assessment is used during a phase of instruction to reinforce learning, so that content is retained.

11. Monitoring learning is the responsibility of all instructional staff and instructional supervisors. Monitoring, including unsatisfactory progress, is done IAW TE SOs or SOPs and CFITES Vol 6 “Conduct of Instruction Programmes”.

12. Continuous monitoring of candidate progress is required to provide the following:

    a. Early warning of trainee difficulties;

    b. Feedback on the effectiveness of training;

    c. Information for any required TRBs; and

    d. Information to assist in completion of the CF 377 Course Report.

13. Trainees experiencing difficulty in any area of performance are to be counselled with regard to the nature of their shortcomings and suggested corrective action IAW TE SOs or SOPs. A record of trainee counselling is to be maintained by the training establishment.

14. Instructional staff shall maintain individual trainee files, including the results of all:

    a. Trainee performance documentation (e.g. Weekly write-up, medical chits, dress inspection chits, conduct chits, etc.);

    b. Formative assessment material (e.g. ECs, practice periods, homework, assignments and quizzes, etc.); and

    c. Summative assessment material (PCs).

TRAINING REVIEW BOARD

15. TRBs shall be convened and conducted IAW with NPTORD 4500-54 “Training Review Board Process”.

16. In the event of prolonged trainee absence, TE Staff shall determine, on a case by case basis and/or iaw the associated QSP, the necessity to convene a TRB.

COURSE REPORTS

17. Course reports (CF 377) shall be prepared and distributed IAW DAOD 5031-9 “Course Reporting and Certificates” and amplified in NPTORD 4500-51 “Course Reports and Certifications”.

18. Trainees will be graded pass or fail. The narrative will indicate the trainee’s strengths and weaknesses based on PC results to provide information to the Career Manager.

19. The Course Report narrative comments could include:

    a. Ability to work with others;

    b. Ability to work under pressure;

    c. Written and verbal communication skills;

    d. Organizational skills; and

    e. Leadership ability.

20. Training limitations and / or incomplete training must be included in the course report.

CHAPTER 3 PERFORMANCE OBJECTIVES AND LESSON SPECIFICATIONS

AUTHORIZATION FOR THIS CHAPTER AND/OR AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER OF THE TRAINING DEVELOPMENT CENTRE

PO 001 Report on ship's power generation and distribution systems (CT0330)

1. Performance. Report on ship's power generation and distribution systems (CT0330).

2. Conditions.

   a. Given:

      (1) Operational power plant, PG&D, and ship;

      (2) References;

      (3) Assistance from Sr Elec Spec;

      (4) IPMS;

      (5) DWAN; and

      (6) DRMIS/WEAF.

   b. Denied: Nil.

   c. Environmental: Any condition onboard ship, at sea or alongside, day or night.

3. Standard. The Elec Sup shall independently report on ship's PG&D systems IAW refs to include:

   a. Advising command on ship's power configuration (PT0010) to include:

      (1) Normal configuration IAW MBS; or

      (2) Risk mitigation COAs for maintaining operational capability when operating abnormal configurations;

   b. Co-ordinating casualty power requirements by:

      (1) Selecting casualty power cable route (ANT0200) IAW A1;

      (2) Plotting BDCS casualty power routes;

      (3) Briefing safety conditions and hazards to casualty power run IC;

      (4) Inspecting casualty power run;

      (5) Verifying casualty power pipe is made; and

      (6) Energizing casualty power run; and

   c. Advising command on ship's PG&D system health to include:

      (1) Equipment deficiencies; and

      (2) Risk mitigation COAs for CM and PM IAW MBS.

4. References. A2, A3, A4, A5, D1.

5. Task, Skill, Knowledge Numbers.
   CT0330, PT0010, and ANT0200.

6. Training Limitations.

a. “Advising command on ship’s PG&D system health” is integral to the performance on the job but is a no train as it is captured in previous training or from experience.

b. Realism of training is not reflective of actual emergency situations to include:

(1) Unable to simulate actual damage to ship; and

(2) Unable to simulate high stress conditions which may occur during action/emergency stations.

PO 001 Report on power generation and distribution systems (CT0330)

EO 001.01 Advise command on ship's power configuration (PT0010).

1. Performance. Advise command on ship's power configuration.
2. Conditions.
   a. Given:
      (1) Operational propulsion control operator trainer (PCOT);
      (2) Scenarios;
      (3) Access to DWAN;
      (4) References; and
      (5) Assistance.
   b. Denied: Nil.
   c. Environmental: Classroom.
3. Standard. The trainee shall independently advise command on ship's power configuration IAW refs by:
   a. Complying with material baseline standards (MBS) IAW A3;
   b. Managing normal power generation configurations based on operational and maintenance requirements by:
      (1) Managing power generation configurations at sea IAW ref A3; and
      (2) Managing power generation configurations alongside IAW ref A3; and
   c. Recommending COAs for mitigation of risk of abnormal configurations at sea, alongside, and with casualty power by:
      i. Analyzing risk;
      ii. Creating COAs; and
      iii. Communicating recommendation to command.
4. Time Required.
   a. EO – 250 mins to deliver training; and
   b. EC – 180 mins (total for all trainees).
5. Remarks. Guest Speaker (MSEO, post-HOD tour) to be organized ahead of time to discuss MBS.
6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LEARNING OBJECTIVE 1: Complying to machinery baseline standard (MBS) by:			

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
1. Explaining machinery baseline standard (MBS) to include: a. Where to access; and b. Applying MBS to electrical readiness.	A3	**Method / Media:** IL, GS. **Trg Environment:** Classroom. **Activity Details:** Trainees will be required to read NAVORD 3000-0 prior to guest speaker IOT be prepared to participate in the discussion. IL to consist of instructor walking the trainees through the various sections of the NAVORD and relating it to their responsibilities on the job. Instructor will ensure all TPts are covered off during the guest speaker and perform confirmation activity. **Trg Resources:** • Handout of NAVORD 3000-0 • Instructor access to DWAN workstation	IL 50 min GS 100 min
LEARNING OBJECTIVE 2: Managing normal power generation configurations based on operational and maintenance requirements			
1. Managing power generation configurations at sea; and 2. Managing power generation configurations alongside.	A2 D1	**Method / Media:** GD. **Trg Environment:** Classroom. **Activity Details:** Training to consist of a guided discussion on the rationale and consequences related to normal power configuration WRT ship's priorities. To include the difference between alongside and at sea configuration considerations. **Trg Resources:** • Guided discussion setup.	GD 50 min
LEARNING OBJECTIVE 3: Recommending COAs for mitigation of risk of abnormal configurations at sea, alongside, and with casualty power			
1. Managing abnormal power generation configurations including: a. Abnormal PG&D configurations at sea to include: i. Decoking; ii. Single switchboard operation; and	A2 D1	**Method / Media:** GD. **Trg Environment:** Classroom. **Activity Details:** Training to consist of a guided discussion on the rationale behind the use and consequences related to abnormal power configurations.	GD 50 min

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
iii. Single DG operation. b. Abnormal configurations alongside. c. Casualty Power.		Focus will be placed on balancing the need for employing abnormal power configurations WRT ship's priorities. To include a brief portion on how casualty power fits into the area of abnormal power configurations. Instructor to ensure trainees are able to analyze risk for creating COAs to brief command. **Trg Resources:** • Guided discussion setup.	

7. Test Details. EO 001.01 will be tested in EC 001.01 and PC 001 IAW table below and Annex K.

EC 001.01: Advise command on ship's power configuration (PT0010)

EC Time: 10 mins + 5 min debrief per trainee (180 min total).

ASSESSMENT ACTIVITY: Trainee will be provided a randomly selected scenario. Scenario will consist of an abnormal ship's power configuration. Trainee will then be required to correctly identify the power configuration (which portion is abnormal and what parts are normal), determine if ship still meets MBS, and provide a simulated brief to command (instructor) outlining multiple COAs. Brief should include the trainee's rationale and risk factors associated with their recommendations.

Pass/fail criteria: Trainees must achieve a SAT on all checklist items to be successful with this assessment.

ASSESSMENT RESOURCES:

- Abnormal power configuration scenario lists.

IMPACT OF FAILURE: Remedial Training / Re-test.

USQ-PO2 Elec Sup – EC 001.01 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1. PASS / FAIL CRITERIA: Trainees must achieve a SAT on all checklist items to be successful with this assessment.
2. IMPACT OF FAILURE: Remedial training, re-test.

ITEM	SAT	UNSAT	COMMENTS
1. Advise command on ship's power configuration			
Identified configuration (normal/abnormal)			
Identified MBS concerns			
2. Communicate ship's power configuration			
Identified operating limitations			
Identified risk factors			
Identified follow up actions			
Provided achievable and practical recommendations based on scenario			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**
THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT	

OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 001 Report on power generation and distribution systems

EO 001.02 Coordinating Casualty Power Requirements.

1. Performance. Coordinate casualty power requirements.
2. Conditions.
   a. Given:
      (1) Operational Portable Operator Unit (POU);
      (2) Scenario involving casualty power;
      (3) Ship's casualty power plan;
      (4) References; and
      (5) Assistance.
   b. Denied: Nil.
   c. Environmental:
      (1) Classroom; and
      (2) PCOT.
3. Standard. The trainee shall independently coordinate casualty power requirements IAW A2, A4, A5 by:
   a. Selecting casualty power cable route (ANT0200) IAW A5 by:
      (1) Utilizing standardized casualty power; and
      (2) Determining alternative sources of casualty power to include:
         i. Rigging direct to DG source; and
         ii. Rigging direct to load (hardwired);
   b. Plotting BDCS casualty power routes by:
      (1) Utilizing pre-determined routes (casualty power lines);
      (2) Utilizing free draw method in BDCS;
      (3) Tasking appropriate SBT/ERT; and
      (4) Updating plot based on continuous reports; and
   c. Inspecting casualty power run to include:
      (1) Power load is isolated and tagged;
      (2) Cables are secured;
      (3) Dangers signs are posted;
      (4) Sentries are posted;
      (5) Sentries are briefed;

(6) Megger™ ground checks are completed;

(7) Load is plugged in;

(8) Source is plugged in;

(9) Casualty power pipe is made;

(10) Casualty power run is energized; and

(11) BDCS is updated.

4. Time Required.

a. EO – 500 mins to deliver training;

b. EC – 60 mins (total for all trainees); and

c. PC – 240 mins (total for all trainees).

5. Remarks. Nil

6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LEARNING OBJECTIVE 1: Selecting casualty power cable routes (ANT0200)			
1. Utilizing standardized casualty power to include: a. Receiving notification from command; b. Selecting route based on damage; and c. Tasking appropriate section base. 2. Determining alternative sources of casualty power to include: a. Receiving notification from command; b. Selecting route based on damage by planning effective route with casualty power IC. c. Rigging abnormal runs to include: i. direct to DG source; and ii. direct to load (hardwired).	A4, A5 part 4 ch 5, A2	**Method / Media:** CS. **Trg Environment:** Classroom. **Activity Details:** Trainees provided with a scenario related to casualty power the day to read the day before the lesson. Instructor to facilitate case study focusing on considerations for configuring casualty power in relation to the scenario. Trainees will gain realistic time appreciation for each portion of the process. **Trg Resources:** • Handouts of scenario	200 min
LEARNING OBJECTIVE 2: Plotting BDCS casualty power routes			
1. Utilizing pre-determined routes (casualty power lines) to include: a. Location of routes in BDCS; and b. How to apply routes in BDCS. 2. Utilizing free draw method in BDCS; 3. Tasking appropriate SBT/ERT. 4. Updating plot based on continuous reports.	A4, A5 part 4 ch 5, A2	**Method / Media:** EDI. **Trg Environment:** PCOT. **Activity Details:** Scenario provided to trainees on day of instruction. Instructor to walk trainees through the scenario, playing the role of command and casualty	200 min

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
		power IC as required. Instructor to explain how to create routes in BDCS. Scenario will consist of a continuation of the one from the previous Learning Objective. Trainees will be required to individually utilize one predetermined route, then one free draw method to properly brief the casualty power IC, updating as required. **Trg Resources:** • Handouts of scenario; and • PCOT.	
LESSON OBJECTIVE 3: Inspecting casualty power run			
1. Inspecting CP cables; 2. Informing command on completed run by requesting permission to energize. 3. Ensuring correct informative pipe from command to include executive order. 4. Energizing CP cables 5. Enforcing safety requirements to include: a. Conducting continuous rounds safety brief; b. Utilizing PPE; c. Using tools (clamp on, ammeter); and d. Performing actions upon discovery of abnormality. 6. Continuous safety pipes IAW SSOs.	A4, A5 part 4 ch 5, A2, CRR 10-11	**Method / Media:** IL **Trg Environment:** Classroom **Activity Details:** Instructor to ensure that the teaching points are covered during the length of the lecture. **Trg Resources:** • Job aid of Pipe for energizing; and • Access to CRR 10-11 to create roundsperson safety brief.	100 Min

7. Test Details. EO 001.02 will be tested in EC 001.02 and PC 001 IAW table below and Annex K.

EC 001.02: Coordinate casualty power requirements

EC Time: 50 mins (concurrent) + 10 min debrief.

ASSESSMENT ACTIVITY: Trainees will be provided a scenario in which they will be required to both identify the type of casualty power, select the correct COA with rationale, and draft the appropriate safety brief to personnel. Trainees will need to plot on BDCS. Assessor will verbally prompt the scenario as required to meet checklist requirements. EC will include questions to meet the assessment criteria for each scenario.

Pass/fail criteria: Trainees must achieve a SAT on all checklist items to be successful with this assessment.

ASSESSMENT RESOURCES:

- PCOT
- Scenario;
- Ship's casualty power drawings;
- Test sheet; and
- Trainee's to provide class developed template.

IMPACT OF FAILURE: Remedial Training / Re-test.

USQ-PO2 Elec Sup – EC 001.02 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1. PASS / FAIL CRITERIA: Trainees must achieve a SAT on all checklist items to be successful with this assessment.
2. IMPACT OF FAILURE: Remedial training, re-test.

ITEM	SAT	UNSAT	COMMENTS
1. Select casualty power cable routes			
Selected optimal route based on damage.			
Identified if route is abnormal or normal.			
2. Plot BDCS casualty power routes			
Utilized pre-existing ship's casualty power drawings (casualty power lines); **OR** Utilized free draw method on paper;			
Indicated which SBT/ERT will conduct the casualty power run			
Updated plot based on "Casualty Power Commenced" report			
Updated plot based on "Casualty Power Energized" report			
Updated plot based on "Casualty Power Completed" report			
3. Brief safety requirements to casualty power IC to include:			
• Circuit			
• Route			
• Hazards			
• Abnormal conditions (if applicable)			
• Mitigation in case of discovery of abnormality			
4. Inspect casualty power run			
Identified a minimum of 5 checks required to inspect within a casualty power run: • Power load is isolated and tagged; • Cables are secured; • Dangers signs are posted; • Sentries are posted; • Sentries are briefed; • Megger™ ground checks are completed; • Load is plugged in; • Source is plugged in;			

ITEM	SAT	UNSAT	COMMENTS
• Casualty power pipe is made; • Casualty power run is energized; and • BDCS is updated			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PC 001: Report On Power Generation and Distribution Systems.

PC Time: 30 mins + 10 min debrief per pair of trainees (240 mins total)

ASSESSMENT ACTIVITY: Trainees will be provided a scenario in which they will be required to complete the following:

- Advise command on ship's power configuration; and
- Coordinate casualty power configurations.

Trainees will provide a brief to command based on their recommendations from the scenario given. Second portion of the assessment will consist of the trainee developing a casualty power plan, utilizing BDCS effectively, and providing a safety brief to casualty power run IC. Briefs will be assessed based on criteria contained in PC checklist and delivered to the assessor.

Pass/fail criteria: Trainees must achieve a SAT on all checklist items to be successful with this assessment.

ASSESSMENT RESOURCES:

- PCOT;
- Casualty power scenario.

IMPACT OF FAILURE: Remedial Training / Re-test / TRB after second unsuccessful attempt IAW NPTORD 4500-54.

USQ-PO2 Elec Sup – PC 001 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1. PASS / FAIL CRITERIA: Trainees must achieve a SAT on all checklist items to be successful with this assessment.
2. IMPACT OF FAILURE: Remedial Training / Re-test / TRB after second unsuccessful attempt IAW NPTORD 4500-54.

ITEM	SAT	UNSAT	COMMENTS
1. Advise command on ship's power configuration			
Identified whether it is a normal or abnormal configuration			
Identified MBS concerns			
2. Communicate ship's power configuration to command			
Identified operating limitations			
Identified risk factors			
Identified follow up actions			
Provided achievable and practical recommendations based on scenario			
3. Coordinate casualty power configurations			
Selected optimal route based on incident			
Identified if route is abnormal or normal			
Justified route choice as optimal and achievable			
4. Plot BDCS casualty power routes			
Utilized pre-existing ship's casualty power BDCS (casualty power lines); **OR** Utilized free draw method on BDCS;			
Indicated which SBT/ERT will conduct the casualty power run; and			
Updated plot based on "Casualty Power Commenced" report			
Updated plot based on "Casualty Power Energized" report			
Updated plot based on "Casualty Power Completed" report			
5. Brief safety requirements to casualty power IC to include:			
• Circuit			
• Route			
• Hazards			
• Abnormal conditions (if applicable)			

ITEM	SAT	UNSAT	COMMENTS
• Mitigation in case of discovery of abnormality			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**
THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT	

OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 002 Advise Command on Ship's Power Isolation

1. Performance. Advise Command on Ship's Power Isolation (PT0200).

2. Conditions.

   a. Given:

      (1) An operational power plant, PG&D, ship;

      (2) References;

      (3) DC organization IAW CFCD 132;

      (4) Functional communications system;

      (5) BDCS;

      (6) Power isolation resources (ex. kill cards); and

      (7) DWAN Access.

   b. Denied: Nil.

   c. Environmental: Any condition onboard ship, at sea or alongside, day or night, during emergency or action stations.

3. Standard. The Elec Sup shall independently advise command on ship's power isolation IAW A5 and A8 by:

   a. Managing Power Isolation (PT0100) by:

      (1) Determining levels of importance to include:

         i. Pre-approved power isolation; and

         ii. Non pre-approved power isolation;

      (2) Interpreting power isolation resources;

      (3) Directing power isolation team by:

         i. Establishing communications with power isolation team to include:

            (i) Maintenance Watch Electrical Technician (MWET);

            (ii) Fwd/Aft switchboard electrician; and

            (iii) ERT HQ; and

         ii. Directing type of power isolation for them to carry out;

         iii. Receiving power isolation team reports;

   b. Briefing DC organization on power isolation requirements by:

      (1) Identifying the event to include:

         i. Flood;

         ii. Fire;

iii. Hazmat;

iv. Collision/Allision; or

v. Other;

(2) Identifying compartment(s) or zone(s) affected by the event;

(3) Analyzing power isolation requirements for the compartment(s) or zone(s) affected by the event to include:

i. MSE equipment considerations;

ii. Mitigation of power isolation impacts to MSE equipment;

iii. Mitigation of power isolation impacts to MBS equipment;

iv. Relationship with command priorities; and

(4) Informing DC organization on circuit isolation intent affecting:

i. CSE equipment;

ii. MBS equipment; and

iii. MSE equipment;

c. Supervising Power Isolation (PT0340) by:

(1) Directing DC organization to isolate power IAW power isolation conference decision;

(2) Tracking power isolation in relation to the dynamic picture of the situation to include:

i. Recording isolated breakers;

ii. Recording tripped breakers; and

iii. Reassessing requirements based on evolving events;

(3) Briefing power isolation team to include:

i. MWET; and

ii. Fwd/aft switchboard electricians;

(4) Communicating status of power isolation by:

i. Making power isolation pipe to include:

(i) Zone isolation;

(ii) Compartment isolation;

(iii) Surgical isolation; and

(iv) No power isolation by Command decision; and

ii. Updating BDCS on status of power isolation;

d. Coordinating reverse power isolation conference by:

(1) Identifying priority of power restoration by:

i. Identifying ship's priority; and

ii. Identifying status of damage as reported by BDAT;

(2) Restore power distribution capabilities by:

i. Analyzing re-energizing considerations to include:

(i) Reparable power distribution capabilities;

(ii) Casualty power considerations; and

(iii) Equipment status;

ii. Directing minor repairs;

iii. Co-ordinating major repairs; and

iv. Determining surgical restorations; and

(3) Briefing power restoration to MSEO to include:

i. MSE equipment considerations; and

ii. Power restoration impact mitigation.

3. References. A1, A2, A4, A6, A7, D1, D2, D3.

4. Task, Skill, Knowledge Numbers. PT0340, PT0100.

5. Training Limitations.

a. Realism:

(1) Will not be able to realistically simulate the stress that occurs during an event.

(2) Cannot simulate the dynamic human interactions and cohesive teamwork that is developed over time.

PO 002 Advise Command on Ship's Power Isolation

EO 002.01 Manage power isolation (PT0100)

1. Performance. Manage Power Isolation.
2. Conditions.
    a. Given:
        (1) Operational PCOT;
        (2) Power isolation book;
        (3) Power isolation scenario;
        (4) References; and
        (5) Assistance.
    b. Denied: Nil.
    c. Environmental:
        (1) Classroom;
        (2) PCOT.
3. Standard. The trainee shall independently manage power isolation IAW A1 by:
    a. Determining levels of circuit importance by:
        (1) Checking validity of pre-approved power isolation list; and
        (2) Checking validity of non-pre-approved power isolation to include:
            i. Vital;
            ii. Semi-vital; and
            iii. Non-vital;
    b. Interpreting power isolation resources by:
        (1) Interpreting kill cards on BDCS; and
        (2) Interpreting power isolation book;
    c. Directing power isolation team by:
        (1) Establishing communications with power isolation team to include:
            i. MWET;
            ii. Fwd/Aft switchboard electrician; and
            iii. ERT HQ;
        (2) Directing type of power isolation for them to carry out; and
        (3) Receiving power isolation team reports;
    d. Briefing DC organization on power isolation requirements by:

(1) Identifying the event to include:

i. Flood;

ii. Fire;

iii. Hazmat;

iv. Collision/allision; or

v. Other;

(2) Analyzing power isolation requirements for the compartment(s) or zone(s) affected by the event to include:

i. MSE equipment considerations;

ii. Mitigation of power isolation impacts to MSE equipment;

iii. Mitigation of power isolation impacts to MBS equipment; and

iv. Relationship with command priorities;

(3) Informing DC organization on circuit isolation intent affecting:

i. CSE equipment;

ii. MBS equipment; and

iii. MSE equipment;

e. Tracking power isolation in relation to the dynamic picture of the situation to include:

(1) Recording isolated breakers;

(2) Recording tripped breakers; and

(3) Reassessing requirements based on evolving events;

f. Briefing power isolation team to include:

(1) MWET; and

(2) Fwd/aft switchboard electricians; and

g. Communicating status of power isolation by:

(1) Making power isolation pipe to include:

i. Zone isolation;

ii. Compartment isolation; and

iii. Surgical isolation; and

(2) Updating BDCS on status of power isolation.

4. Time Required.

a. EO – 600 mins to deliver training; and

b. EC – 130 mins for all trainees.

5. Remarks.

a. Training should commence with simple scenarios, gradually increasing in complexity as the trainees' skills improve.

b. Guest speaker (command qualified NWO) to be arranged ahead of time.

6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE 1: Determining levels of circuit importance			
1. Determining levels of importance for Pre-approved power isolation by: a. Checking validity of pre-approved power isolation list; and b. Drafting COTM related to pre-approved power isolation. 2. Determining levels of importance for Non pre-approved power isolation to include: a. Vital; b. Semi-vital; and c. Non-vital.	A1 A4 A5 A6 D1 D2 D3	**Method / Media:** IL; homework assignment. **Trg Environment:** Classroom. **Activity Details:** Instructor to provide trainees with an example of a pre-approved isolation list. Instructor is to walk trainees through the process of verifying the validity of the example. Instructor to walk the trainees through the rational of why certain systems are considered vital, semi-vital, and non-vital. Provide the trainees with a homework assignment in which they are required to determine if certain pieces of kit are vital, semi-vital, or non-vital. Assignment will be reviewed in class. **Trg Resources:** • Examples ship's specific COTM; and • Homework assignment.	40 min 10 min review
LESSON OBJECTIVE 2: Interpreting power isolation resources			
1. Interpreting kill cards on BDCS. 2. Interpreting power isolation book.	A1 A4 D2 D3	**Method / Media:** EDI. **Trg Environment:** PCOT. **Activity Details:** Trainees are familiar with the concept of kill cards, however lack the necessary knowledge of the content required for an electrical supervisor. Instructor will lead the trainees through the processes of interpreting the information on the kill cards to brief the MSEO. Trainees to be provided with various scenarios where they will	100 min

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
		be required to utilize the information contained in the kill cards to identify isolation effects on select pieces of equipment. **Trg Resources:** • Operational PCOT; • Example of ship's power isolation book; • Scenarios.	
LESSON OBJECTIVE 3: Directing power isolation team			
1. Establishing communications with power isolation team to include: a. Maintenance Watch Electrical Technician (MWET); b. Fwd/Aft switchboard electrician; and c. ERT HQ; and 2. Directing type of power isolation for them to carry out; 3. Receiving power isolation team reports	A1 A4 A5 A7 D2 D3	**Method / Media:** GD **Trg Environment:** Classroom **Activity Details:** Instructor to facilitate a guided discussion on directing the power isolation team. Discussion points should draw on previous experience from the trainees and be focused on the new perspective of the electrical manager. **Trg Resources:** • Guided discussion setup.	50 Min
LESSON OBJECTIVE 4: Identifying the event			
1. Identifying power isolation requirements for: a. Flood to include: i. Immediate actions; and ii. Time considerations. b. Fire to include: i. Location considerations; ii. Immediate actions; and iii. Follow on actions. c. Hazmat by crashing power and ventilation. d. Collision/Allision to include; i. Location of damage; ii. Extent of damage; and iii. Damage mitigation.	A1 A4 A5 D1	**Method / Media:** GD. **Trg Environment:** Classroom. **Activity Details:** Instructor to facilitate a guided discussion related to the various factors that influences the electrical manager's response to events at sea. Discussion will capitalize on trainees' previous experiences. Instructor is to emphasize that immediate actions required are critical to successful power isolation. Possible inclusion of previously used sea training scenarios. **Trg Resources:** • Guided discussion setup.	100 min
LESSON OBJECTIVE 5: Analyzing power isolation requirements for the compartment(s) or zone(s) affected by the event			
1. Determining impact of isolating circuits on CSE equipment to include:	A1 A3	**Method / Media:** Self-Study assignment; IL	50 min

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
a. Identifying circuits; and b. Passing information to ERT HQ. 2. Determining impact of isolating circuits on MBS equipment to account for safe at sea requirements. 3. Determining impact of isolating circuits on MSE equipment to include: a. Propulsion impact; and b. Power generation considerations.	A4 A5 A7 D2 D3	**Trg Environment:** Classroom. **Activity Details:** Trainees are to be provided a reading assignment NLT two days prior to delivering lecture. Assignment to consist of a series of questions related to determining the impact of isolating circuits. Instructor is to correct reading assignment in class while ensuring the teaching points contained in this lesson are covered during the lecture. **Trg Resources:** • Reading assignment.	
LESSON OBJECTIVE 6: Identifying power isolation requirements in relation to command priorities			
1. Identifying degrees of readiness priorities. 2. Operating within ship's priorities to include: a. Float; b. Move; and c. Fight.	A1 A2 A4 A5 D1	**Method / Media:** IL with GS **Trg Environment:** Classroom. **Activity Details:** Instructor will ensure the teaching points contained in this lesson are covered during the lesson by the guest speaker. Focus of the lecture will be familiarization of the degrees of readiness and how the ship's priorities will impact power isolation. Guest lecturer should be a command qualified officer who will be able to explain how command determines their priorities during events. **Trg Resources:** • Guest Speaker- Command Qualified NWO.	100 min
LESSON OBJECTIVE 7: Tracking power isolation in relation to the dynamic picture of the situation			
1. Informing CSE of intent to isolate power. 2. Recording isolated breakers. 3. Recording tripped breakers 4. Reassessing requirements based on evolving events.	A1 A4 A5 D2 D3	**Method / Media:** GD. **Trg Environment:** Classroom. **Activity Details:** Guided discussion to consist of the importance of informing CSE on up-coming isolation and keeping an up to date account of all isolated breakers. Discussion should include re-assessing previous considerations for	50 min

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
		isolation of space(s) based on updated warfare priority. **Trg Resources:** • Guided discussion setup.	
LESSON OBJECTIVE 8: Briefing power isolation team			
1. Briefing a relocation plan for the following: a. MWET relocation b. Switchboard relocation c. MCR / HQ1 relocation 2. Re-establishing communication when lost 3. Briefing MWET to include: a. Role and responsibilities to include; i. Employment. b. Identifying breakers; c. Confirming breakers; and d. De-energizing breakers. 4. Briefing Fwd/aft switchboard electricians to include: a. Role and responsibilities to include; i. Employment. b. Identifying breakers; c. Confirming breakers; and d. De-energizing breakers.	A1 A4 A5 A6 A7 D2 D3	**Method / Media:** GD **Trg Environment:** Classroom **Activity Details:** Instructor to lead a guided discussion on the employment of MWET and Fwd/aft switchboard electricians. Discussion will be based on experience as no standing policy exists. Instructor to emphasize the importance of having a relocation plan in place and the requirement to re-establish communications with the electrical supervisor. **Trg Resources:** • Guided discussion setup.	50 min
LESSON OBJECTIVE 9: Communicate status of power isolation			
1. Identifying correct power isolation pipe from the following: a. Zone isolation; b. Switchboard isolation; c. Compartment isolation; d. Surgical isolation; and e. No power isolation by Command decision. 2. Reporting on power isolation by: a. Broadcasting appropriate informative pipe; and b. Briefing MSEO to include concurrence with battle rhythm pipe.	A1 A4 A5 D2 D3	**Method / Media:** Table top / Walkthrough **Trg Environment:** Classroom **Activity Details:** Instructor to ensure teaching points are covered during the course of the lesson. Activity will consist of a table top exercise in which trainees will be presented scenarios related to power isolation including a BDCS event overview. Trainees will then be required to compare current power isolation status to battle rhythm pipes from the executive officer, indicating their concurrence or non-concurrence. **Trg Resources:**	50 min

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
		• Scenarios; and • Battle rhythm pipe script.	

7. Test Details. EO 002.01 will be tested in EC 002.01 and PC 002 IAW Annex K.

EC 002.01: Manage Power Isolation

EC Time: 120 mins (concurrent) + 10 min debrief.

ASSESSMENT ACTIVITY: Trainees will be provided an event scenario of a fire in a space requiring power isolation. The compartment will have both MSE and CSE considerations at a minimum. Trainees will complete the assessment using PCOT and written sheets. Trainees will have to select the correct kill card/isolation book section and use the information it contains to determine power isolation recommendations. These recommendations shall be based on the event and include:

- Pre-approved vs non-pre-approved
- Impact on isolated equipment.

Trainees will then continue the scenario requiring them to determine CSE/MSE/MBS power isolation considerations while verbally communicating effectively with ERT HQ (if required) and MSEC. To communicate with assessor, trainee will raise their hand when ready to present their considerations. Immediate actions will also be a component of this assessment.
Trainees will have to provide the content for a brief to the DCO, including mitigation options based on information in the scenario (the scenario may be broken into sections and handed out at time of need if required). A relocation plan for MWET and switchboard personnel and responses to battle rhythm pipes will also be developed by the trainees.

Pass/fail criteria: Trainees must achieve a SAT on all checklist items to be successful on this assessment.

ASSESSMENT RESOURCES:

- PCOT;
- Power isolation resources (kill cards) pre-loaded;
- Battle rhythm pipe samples;
- Briefing template; and
- Scenario.

IMPACT OF FAILURE: Remedial Training / Re-test.

USQ-PO2 Elec Sup – EC 002.01 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1. PASS / FAIL CRITERIA: Trainees must achieve a SAT on all checklist items to be successful on this assessment.
2. IMPACT OF FAILURE: Remedial Training / Re-test.

ITEM	SAT	UNSAT	COMMENTS
1. Manage Power Isolation			
Identified pre-approved breakers			
Identified immediate actions required for pre-approved breakers			
Identified non-pre-approved breakers (CSE, MBS, MSE)			
Described reasoning for why they are non-pre-approved based on MBS and command priorities			
Recommended power isolation requirements and mitigations for non-pre-approved MSE breakers			
Identified scope and scale of event(s)			
Communicated affected circuits to ERT HQ			
Described impact on propulsion and power generation			
Communicated affected circuits to MSEC			
Described impact on ship's priorities			
2. Track power isolation in relation to the dynamic picture			
Recorded isolated / tripper breakers			
Reassessed requirements based on scenario events			
3. Brief power isolation team			
Briefed MWET / MCR / HQ1 relocation plan			
Briefed Fwd/aft switchboard relocation plan			
4. Communicate status of power isolation			
Justified whether battle rhythm pipe was correct or incorrect			
Updated power isolation log			
Updated BDCS on status of power isolation			

ADDITIONAL COMMENTS

STRENGTHS:		AREAS FOR DEVELOPMENT:	
THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 002 Advise Command on Ship's Power Isolation

EO 002.02 Coordinate reverse power isolation conference

1. Performance. Coordinate reverse power isolation conference.
2. Conditions.
   a. Given:
      (1) Operation PCOT;
      (2) Power isolation book;
      (3) Power isolation scenario;
      (4) References; and
      (5) Assistance.
   b. Denied: Nil.
   c. Environmental:
      (1) Classroom; and
      (2) PCOT.
3. Standard. The trainee shall coordinate reverse power isolation conference IAW CFCD 132-A-8-1 by:
   a. Identifying priority of power restoration by:
      (1) Identifying ship's priority; and
      (2) Identifying status of damage as reported by BDAT;
   b. Restore power distribution capabilities by:
      (1) Analyzing re-energizing considerations to include:
         i. Reparable power distribution capabilities
         ii. Casualty power considerations; and
         iii. Equipment status;
      (2) Directing minor repairs;
      (3) Co-ordinating major repairs; and
      (4) Determining surgical restorations; and
   c. Briefing power restoration to MSEO to include:
      (1) MSE equipment considerations; and
      (2) Power restoration impact mitigation.
4. Time Required.
   a. EO – 150 mins to deliver training;
   b. EC – 60 mins (total for all trainees); and

c. PC – 220 mins (total for all trainees).

5. Remarks. Nil.

6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE 1: Identify priority of power restoration			
1. Identifying hazards of re-energization including purpose of re-energization conference 2. Identifying ship's priority to include recognizing requirement to achieve CSE/Deck/MSE priorities. 3. Identifying status of damage as reported by BDAT.	A1 A3 A4 A7 D2 D3	**Method / Media:** IL. **Trg Environment:** Classroom. **Activity Details:** Instructor to ensure all teaching points are covered during the lesson. The requirement to have circuit numbers provided by CSE department is to be emphasized, along with the importance of concurring with command's intent. Anticipated responses from BDAT are to be discussed with appropriate examples provided. **Trg Resources:** • Lecture PPt.	50 min
LESSON OBJECTIVE 2: Restore power distribution capabilities			
1. Analyzing re-energizing considerations based on BDAT survey to include: a. Reparable power distribution capabilities b. Casualty power considerations; and c. Equipment status; 2. Directing minor repairs; 3. Co-ordinating major repairs; 4. Determining surgical restorations;	A1 A3 A4 A7 D2 D3 Various repair manuals appropriate to equipment	**Method / Media:** IL, GD. **Trg Environment:** Classroom. **Activity Details:** Instructor to ensure that all material in teaching points is covered during the lecture. Guided discussion is to focus on the interaction between the BDAT, electrical supervisor, other departments. Process for developing a power restoration plan to be discussed in class with trainees capitalizing on their previous experience. **Trg Resources:** • Guided Discussion set up; and • IL ppt.	50 min

LESSON OBJECTIVE 3: Brief power restoration to MSEO			
1. Briefing power restoration considerations for MSE equipment to MSEO 2. Briefing power restoration considerations for CSE equipment to CSEO 3. Mitigating power restoration impact.	A1 A3 A4 A7 D2 D3	**Method / Media:** IL **Trg Environment:** Classroom. **Activity Details:** Instructor to ensure that all material in teaching points is covered during the lecture. Trainees receive template of the EO brief and short scenario. They will be required to complete the template using the information contained within the scenario. Instructor will randomly select trainees to provide their brief to the class. **Trg Resources:** • IL ppt; • MSEO briefing template.	50 min

7. Test Details. EO 002.02 will be tested in EC 002.02 and PC 002 IAW table below and Annex K.

EC 002.02: Coordinate Reverse Power Isolation Conference

EC Time: 50 mins (concurrent) + 10 min debrief.

ASSESSMENT ACTIVITY: Based on the scenario from EC 002.01, trainees will be provided the next stage in the scenario and will be required to coordinate reverse power isolation conference. The scenario will include reports from BDAT, ERT HQ, and written pipes stating ship's repair priority. Trainees will use this material to develop a written restoration plan for power accounting for all pertinent considerations.

Pass/fail criteria: Trainees must achieve a SAT on all checklist items to be successful on this assessment.

ASSESSMENT RESOURCES:
- NAVORD 3000-0 (MBS)
- Sample BDAT report; and
- Sample communication from ERT HQ.

IMPACT OF FAILURE: Remedial Training / Re-test.

USQ-PO2 Elec Sup – EC 002.02 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1. PASS / FAIL CRITERIA: Trainees must achieve a SAT on all checklist items to be successful on this assessment.
2. IMPACT OF FAILURE: Remedial Training / Re-test.

ITEM	SAT	UNSAT	COMMENTS
Analyze Reverse Power Isolation Conference by:			
Identified ship's priority			
Described the concerns with CSE equipment circuit identification			
Identified the repair priority based on BDAT survey			
Identified other departments' requirements			
Briefed MSEO on power restoration plan to include mitigating power restoration impact			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**
THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT	

OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PC 002: Advise Command on Ship's Power Isolation

PC Time: 200 mins (concurrently) + 20 mins debrief.

ASSESSMENT ACTIVITY: Trainees shall be given a scenario involving a single significant incident (fire, flood, battle damage, etc.). Individually, trainees will using information provided to advise command on ship's power isolation. This will consist of utilizing BDCS and kill cards.

Scenario to include (Divided into four sections):

- Event with both CSE/MSE considerations
- Sample pipes
- Sample report from switchboard
- Report from EOOW
- Sample BDAT report
- Sample communication from ERT HQ

Pass/fail criteria: Trainees must achieve a SAT on all checklist items to be successful with this assessment.

ASSESSMENT RESOURCES:

- NAVORD 3000-0
- Scenario
- PCOT
- Power isolation resource
- References

IMPACT OF FAILURE: Remedial Training / Re-test / TRB after second unsuccessful attempt IAW NPTORD 4500-54.

USQ-PO2 Elec Sup – PC 002 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1. PASS / FAIL CRITERIA: Trainees must achieve a SAT on all checklist items to be successful on this assessment.
2. IMPACT OF FAILURE: Remedial Training / Re-test / TRB after second unsuccessful attempt IAW NPTORD 4500-54.

ITEM	SAT	UNSAT	COMMENTS
1. Manage Power Isolation			
Identified pre-approved breakers			
Identified immediate actions required for pre-approved breakers			
Identified non-pre-approved breakers (CSE, MBS, MSE)			
Described reasoning for why they are non-pre-approved based on MBS and command priorities			
Recommended power isolation requirements and mitigations for non-pre-approved MSE breakers			
Identified scope and scale of event(s)			
Communicated affected circuits to ERT HQ			
Described impact on propulsion and power generation			
Communicated affected circuits to MSEC			
Described impact on ship's priorities			
2. Track power isolation in relation to the dynamic picture			
Recorded isolated / tripper breakers			
Reassessed requirements based on scenario events			
3. Brief power isolation team			
Briefed MWET / MCR / HQ1 relocation plan			
Briefed Fwd/aft switchboard relocation plan			
4. Communicate status of power isolation			
Justified whether battle rhythm pipe was correct or incorrect			
5. Analyze Reverse Power Isolation Conference			
Identified ship's priority			
Described the concerns with CSE equipment circuit identification			
Identified the repair priority based on BDAT survey			
Identified other departments' requirements			

ITEM	SAT	UNSAT	COMMENTS
Briefed MSEO on power restoration plan to include mitigating power restoration impact			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

ANNEX A - *MAIN REFERENCES*

A = CF or other official Canadian government publication

B = Official Allied Force or Allied government publication

C = Commercial publication

D = Other

Ref Code	Publication
A1	CFCD 132-A-8-1 (HFX Power isolation procedure)
A2	SSOs
A3	NAVORD 3000-0, Materiel Baseline Standard (MBS) - Surface Ships-Policy
A4	CFCD 102 Combat Readiness Requirements
A5	CFCD 132 RCN Damage Control Manual
A6	D03003005/SF000 General Electric specification for Canadian Forces Ships. Ch. 7 and Ch. 9
A7	SEMS
A8	CFCD 133 Emergency Repair Team Manual
D1	Ship's specific COTM/CSO/SOP
D2	Ship's power isolation book
D3	Hfx Class Kill Cards

ANNEX B - *TASK LIST*

Task No	Task Statement	Train / No Train	PO
CT0330	Report On Ship's Power Generation And Distribution Systems	T	001
PT0010	Advising Command On Ship's Power Configuration	T	001
ANT0200	Configuring Casualty Power Cable Routes	T	001
PT0100	Manage Power Isolation	T	002
PT0340	Supervise Power Isolation	T	002
IT0810	Surveying battle damage	NT	NIL
IT0120	Assessing battle damage to engineering systems	NT	NIL
AFT1520	Investigating major equipment failures	NT	NIL
PT0090	Lead Collective Operation of Power Generation and Distribution System	NT	NIL
AFT1510	Inspect Completed Repair or Installation Work	NT	NIL

ANNEX C - *QUALIFICATION STANDARD AND PLAN WRITING BOARD RECORD OF PROCEEDINGS*

Dates: 11 April 2018 – 2 May 2018

Location: Workpoint ATCO 112

Board Members:

Position	Rank and name	MOSID / Unit	Dates
TM	CPO2 Gagné	00379/NPTGHQ	11 April 2018 – 2 May 2018
Chair	CPO2 MacDonald	00379/STP(P) SUB	11 April 2018 – 2 May 2018
SME	PO1 Penney	00379/NTDC(P)	11 April 2018 – 2 May 2018
SME	PO1 Huber	00379/NTDC(P)	11 April 2018 – 2 May 2018
SME	PO2 Gillies	00379/NTDC(P)	11 April 2018 – 2 May 2018
TDO	SLt Jong Kook Lee	00211/NTDC(P	11 April 2018 – 20 Apr 2018
TDO	Lt(N) Todd Robinson	00211/NTDC(P)	11 April 2018 – 2 May 2018
Secretary	Sunny Sihota	Civilian	11 April 2018 – 2 May 2018

Board Objective:

The purpose of this writing board was to develop the QSP for the USQ PO2 Electrical Supervisor in the overall context of the Marine Technician trade development MIP. This was the eighth QSPWB to be convened in the process.

Discussions / Decisions:

**Decisions which are stricken through were changed at a later date*

Date	Decision/Event	Detail
16 APR 18	Intro Brief	TDO lead the board through the intro presentations into the board process, CFITES, and roles and responsibilities of the board members. Conducted document review with SMEs to include: JTAR, Occ spec, training progression.
	Document Review	During our review of occupation specifications it was determined that of the initial two points of electrical supervisor were a cut and paste from previous electrician trade. These two points need to be moved to master seaman USQ to more accurately reflect current mar tech electrical supervisor duties.
	Task Grouping	The board commenced grouping of tasks. Once this was completed the TDO led the board through review of tasks determined to be NO TRAINED by the JTAR
	AFT1510	Inspect completed repair or installation work. NT expected to pick up w EXP
	ANT0200	Configure casualty power cables Routes NT Gained with EXP
	PT0090	Lead collective operation of power generation and distribution system NT gained with EXP

Date	Decision/Event	Detail
	Scalaring	TDO led SMEs through scalaring process for the tasks. Subject matter experts commenced scalaring tasks in order of job performance
	BDAT	Damage controls scenarios provided by sea training should allow time for BDAT to be conducted
	Definition	The board determined when referencing the word ship it is to include all manner of sea going vessels, such as CPF, MCDV and Submarine.
	Performance Objectives	The TDO led the board through the process of drafting POs for the PO2 Electrical Supervisor.
	CPO2 Harel	Sea training sr electrician CPO2 Harel spoke with the board after reviewing the scalared content and had the following concerns: 1. BDAT falls under the control of the CSE Department (HFX). 2. Ship's staff have been deficient in their work preparing for sea training evolutions. Particular attention should be paid on the preps required to be completed and submitted to STP prior to workups.
	CPO2 Gagné Cdr Sauve	Training manager briefed the board and reconfirmed the position of the RQ PO2 and PO1 in the training scheme. The commanding officer conducted his welcoming brief to the board and reconfirmed his mandate. Discussion led to the content and scheduling of the MS Electrician USQ and how it should be a controls course followed by modularized training package that was platform specific. Board members felt that training time was being wasted having IPMS being repeated on both the legacy QL 6a and IPMS technician course. Once the Commanding Officer departed CPO2 Gagné then discussed with the board their recommendations for the training of the MS.
17 APR 18	Scalaring	While scalaring the tasks for *readiness preps for workups* it was discovered that personnel management had not been addressed in the PO1 or PO2 common RQ's. This necessitated its addition into the PO2 RQ common course as the board felt that this was the first stage in the electrical progression where it was required to be instructed. The board would confer with the concurrently running PO2 Mechanical board to receive their input.
	INELP	The board discussed the lack of personnel management and divisional administration training present at the PO2 and PO1 RQ QSPs. Legacy QL6b course contained a portion in PO 002 and 003 which addressed these concerns. The board will review this legacy section for possible inclusion in the PO2 RQ to address this deficiency with the recommendation that it be included into the INELP program or PO2 RQ where appropriate.
	Target population analysis	TDO led the board in the target population analysis with the board attempting to determine the characteristics of the individuals who would eventually be trained on the course. The board felt that the candidates would have a high level of technical knowledge as well as a familiarity with technology. Trainees with a large potential range in formal education would also be a consideration.
	CPO2 Gagné	TM met with the board to discuss the scheme of training for MS Electrician course that was brought up yesterday. Board determined that the two week class package for Halifax class could be taught in house on each coast. Kingston class vessels class package could be accomplished as an OJT of two weeks in duration. Additionally, the TM indicated that INELP was not a mandatory course for MAR TECH. As a result, IOT capture the essential teaching points previously contained in PO 002 and 003 of the legacy QL 6b Electrician course it should be moved to the PO2 RQ QSP.
18 APR 18	Gridlock	PO2 Mechanical board was invited to discuss the merits of the inclusion of portions of the QL 6b electrical legacy QSP. Specifically PO 002 TP f.5 related to tiered readiness programs and PO 003 departmental maintenance

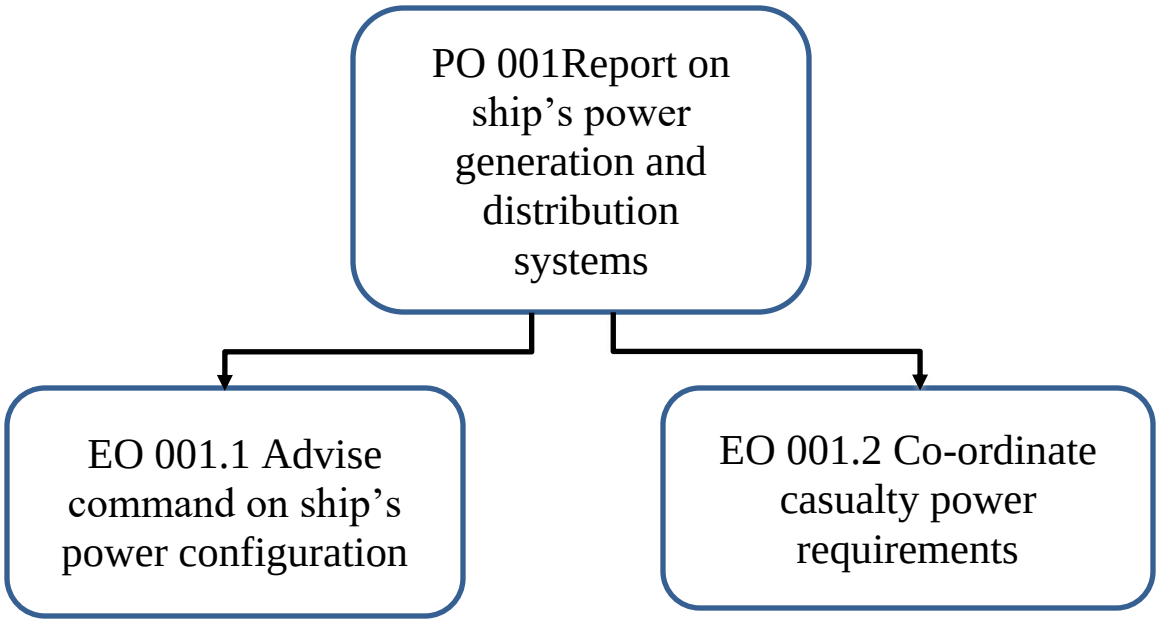
Date	Decision/Event	Detail
		were discussed. It was unanimously decided that the content of PO 003, once updated to account for recent changes in management programs was to be incorporated in the PO2 RQ course. The content of EO 002.02 F.5 was determined to be performed at the PO1 RQ level.
	New Trade Progression	The PO2 Mech and PO2 Elec board members agreed on including stability and personnel management content (aforementioned) in the PO2 Common course. The Mech and Elec portions will be separate USQs immediately following the PO2 RQ Common.
	Instructional Analysis	The board continued with instructional analysis related to the issue of power isolation.
	ANT0200	Configure casualty power cables routes task was revisited and deemed now to be trained in the current context of the instructional analysis process.
19 APR 18	Meeting with CNPTG	Board chair and TDO attended a teleconference with MSE Div commander LCdr Day and TM CPO2 Gagné. Discussion centered on the movement of portions of the QL 6b electrical legacy QSP into the RQ PO2 and RQ PO1 QSPs. CNPTG agreed with the content move WRT the electrical content and additionally directed the re-allocation of EO 004.01 and EO 004.03 into the USQ PO2 electrical supervisor course. Additionally, it was determined that the board would, once they completed the PO2 USQ would design the USQ MS Electrical QSP based on a common controls course, IPMS module, and Kingston class module(if required).
	PO1 RQ	Removed content from EO 004.01 and 03 and repositioned it within the USQ PO2 Electrical Supervisor QSP.
	Instructional Analysis	With the inclusion of the new material from the RQ PO1 QSP, the board resumed the process of instructional analysis on the remaining tasks/knowledge/skills/attitudes.
20 APR 18	IT0120	Assessing battle damage to engineering systems (IT0120) the board felt that this should also be applied to all methods of damage to include but not limited to allisions, collisions, and failures.
	PO 003	Board completed the design for the material moved to the PO2 USQ from the PO1 RQ.
	MBS	The board determined that having a guest speaker come in to speak with the trainees during the MBS portion would be extremely useful both in imparting knowledge and establishing a face to face connection.
23 APR 18	Instructional Design	The board continued where they left off the previous week, finalizing the assessment of PO 001. It was decided that the assessment of the PC could take place in both the PCOT and BDCS trainer, allowing two trainees to be assessed at one time. This would require an additional assessor to be present. The board then transitioned to designing the final remaining Performance Objective.
24 APR 18	Instructional Design	The board determined that the use of the word “Event” would indicate all manner of incidents such as flood, fire, collision, allision, hazmat, etc that would necessitate a response from the electrical supervisor.
	References	CFCD 132-A-8-1 clearly describes the procedure to isolate power. This reference shall be used as the primary means for instructors to describe the process to trainees as opposed to injection of personal experience.
	Learning Objectives	The board discussed re arranging the learning objectives from EO 002.02 to better reflect the process flow which occurs during an event with regards to determining priorities for the electrical supervisor.
25 APR 18	Instructor requirements	The board discussed the requirement of the course instructor being post PO1 electrical manager. This would ensure that the instructor would have real life experience in the processes that they will be providing instruction.

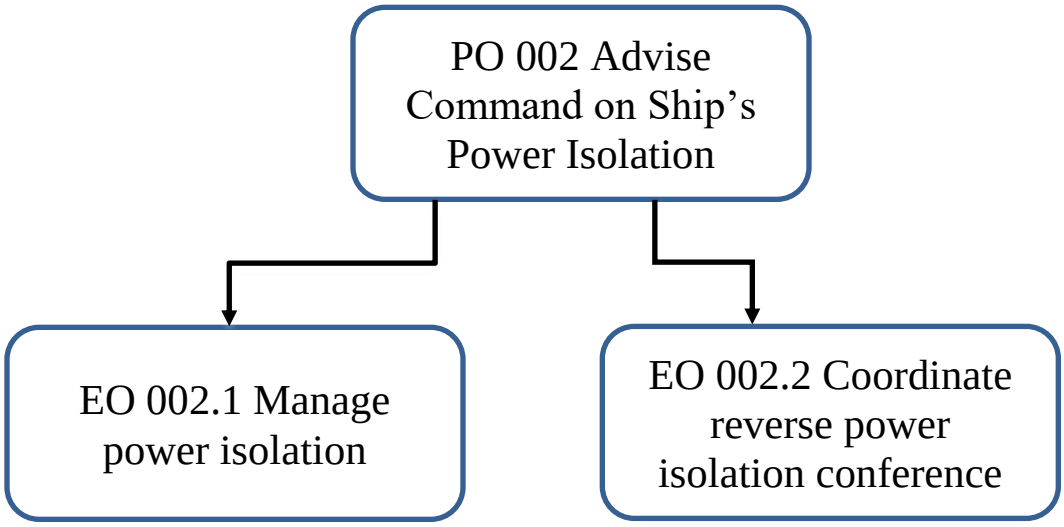
Date	Decision/Event	Detail
	Sea training	The usage of previous sea training event scenarios was determined by the board to be a valuable training resource. Contact with sea training should be made to access this resource.
	High voltage	The board determined that high voltage training would be delivered as just in time training and would not be covered in this QSP as it currently exists.
	MWET	Roles and responsibilities of the MWET during action/emergency stations should be defined by the occupation. This prevents any uncertainty as to the capabilities of this member when they get to a ship. At a minimum, this individual should be legacy ETECH ETOW qualified. (Main concern is after legacy ETOW are no longer within the fleet).
26 APR 18	Policy Creation	The board felt that policy needed to be created that standardized the actions of the MAR TECH, manager Elec during action/emergency station. The board designed the QSP material using best practice and personal experience, but feel as the MAR TECH trade matures SOPs and policy need to be developed.
	CSE ERT HQ	Communication with regards to power isolation needs to be standardized. This is required to prevent miscommunication between ERT HQ and DC HQ. For example DC HQ communicated using circuit designators, EG. F476. ERT HQ communicated using equipment names, e.g. SG-150. This is fine when speaking with each organization, however it causes confusion when dealing with outside departments.
	Board Finalization	The board expressed disappointment with the constantly changing direction and taskings provided to the board. They felt this detracted from the focus of the board and led to frustrations.

Follow Up Action:

Item	Action	OPI	Expected Date of Completion
1	usage of previous sea training event scenarios		
2	Roles and responsibilities of the MWET during action/emergency stations		
3	high voltage training		
4			
5			
6			

ANNEX D - *QSPWB SCALAR*





ANNEX E - *ON JOB PERFORMANCE RECORD (OJPR)*

There is no OJT or OJPR component to this training.

ANNEX F - *ACRONYMS AND ABBREVIATIONS*

Acronym / Abbreviation	Term or Definition
BDAT	Battle damage assessment team
BDCS	Battle damage control system
CM	Corrective Maintenance
COA	Course of Action
COTM	Commanding Officer's Temporary Memorandum
CP	Casualty power
CS	Case Study (instructional strategy)
CSE	Combat Systems Engineering
CSEO	Combat Systems Engineering Officer
CSO	Captain's Standing Orders
DC	Damage Control
DG	Diesel Generator
EDI	Explain-Demonstrate-Imitate (instructional strategy for skill lessons)
EOOW	Engineering Officer of the Watch
ERT	Emergency Repair Team
GD	Guided discussion (instructional method)
GS	Guest speaker (instructional media)
HQ	Headquarters
IC	In charge
JTAR	Job task analysis report
MBS	Machinery Baseline Standards
MSEC	Marine Systems Engineering Chief
MSEO	Marine Systems Engineering Officer
MWET	Maintenance Watch Electrical Technician
PCOT	Propulsion Control Operator Trainer
PG&D	Power Generation and Distribution System
PM	Preventative Maintenance
POU	Portable Operator Unit
SBT	Section Base Team
SOP	Standard Operating Procedures

ANNEX G - *SUGGESTED SEQUENCING AND TIMINGS*

1. This annex is to be used by Campus Support to properly schedule all required classrooms, trainers, ships, personnel, etc for the duration of the course. Admin and PT periods are accounted for IAW NPTORD 4500-22 Annex B (QSP Instructions).
2. The tables are completed in the preferred sequence for lessons and resources.

EO ###.##	EO Performance Statement	Personnel resources	Required Resources (for scheduling)	Trg Time per resource (mins)	EC Time (mins)	PC Time (mins)
PO 001: Report on ship's power generation and distribution systems				Total PO time:		1290
PO ATT:	1230	PO Admin Time:	60	PO PT time:	0	
EO 001.01	Advise command on ship's power configuration	Instructor	Classroom w/ DWAN	150	180	
		Guest speaker - MSEO Post HOD tour	Classroom w/ DWAN	100		
EO 001.02	Coordinate casualty power requirements	Instructor	Classroom w/ DWAN	300		
		Instructor	PCOT	200	60	240
PO 002: Advise Command on Ship's Power Isolation				Total PO time:		1220
PO ATT:	1160	PO Admin Time:	60	PO PT time:	0	
EO 002.01	Advise command on ship's power configuration	Instructor	Classroom w/ DWAN	400		
		Instructor	PCOT	100	130	
		Guest speaker - Command Qualified NOW	Classroom w/ DWAN	100		

EO 002.02	Coordinate reverse power isolation conference	Instructor	Classroom w/ DWAN	150	60	220

ANNEX H - *ADDITIONAL TRAINING RESOURCES*

In accordance with Chapter 1 para3-4, add any extra tables or paragraphs used for training resources in this annex. (**Annex H not to be deleted**)

1. PCOT requirements as of 2022:

   a. Room to include:

      (1) Ventilation to L3 standard
      (2) Power source to L3 standard
      (3) 2 whiteboards, mounted

   b. 12 trainee stations that each include:

      (1) 1 desktop computer
      (2) 2 monitors
      (3) Mouse and keyboard
      (4) Desk to house IT resources
      (5) Chair

   c. Instructor station that includes:

      (1) 1 desktop computer
      (2) 4 monitors
      (3) Mouse and keyboard
      (4) Desk to house IT resources
      (5) Chair
      (6) Projector with screen

   d. PCOT software for each station to include:

      (1) IPMS network
      (2) OBTS
      (3) PCOT Initial Conditions
      (4) Orchid software
      (5) MS OS

   e. L3 Field Service Representative support

   f. PCOT / EHM server to include:

      (1) Server cabinet
      (2) Uninterrupted Power Supply (UPS)
      (3) Server rack, display, and keyboard

ANNEX I - *COSTING INFORMATION*

Include information about cost in whichever format is applicable (**Not to be deleted**).

Estimates Template based on CF Cost Factors Manual FY 2012/13

Template Annex I Costing.xls

****5 year total based on one course per year INCLUDING Initial Set-Up. Costs may be adjusted accordingly.**

Course:		Serial:				
TD	No. Of Students / instructors /	Cost per each	# Days (if applicable)	Initial TE Set-up	Each successive course	5 Year total
Subtotal						
Material	Quantity	Cost per each	# Days (enter 1 if not applicable)			
Subtotal						
PYs (only if course includes incremental on SWE, or course to be delivered off site	Quantity	Cost per day* (SWE if necessary)	# Days (if applicable)			
	Total Estimated Costs					

ANNEX J - *COURSE HISTORY*

This annex shall NOT be deleted. It must be annotated with every update or modification to this QSP. This is in addition to making the appropriate entry in the Record of Amendments table. Details on course history issues such as explanations of amendments, pertinent findings, trends, resource problems and modifications shall be captured in the table below. Focus should be given to both excellent and questionable issues of the course. Capturing important course history information is a useful managerial tool for amendment considerations by future course developers.

Reason for amendment (with date)	Details on course history
QSPWB (11 April 2018 – 2 May 2018)	The MAR TECH USQ PO2 Electrical supervisor course was designed using elements from the pre-existing legacy QL6b course. Sea training was invited to address the board in order to help the board identify and address any observed deficiencies in the fleet for this position.
QSP Review 28-29 Apr 22	PO2 Alex Moss and PO1 Brendan Huber with Learning Designer Sarah Valentine collaborated to ensure items found in QA reviews is fixed to be able to send the QSP for approval.
	Discussion on whether the material within this training is for an Electrical Supervisor or the Electrical Manager. 1. This training comes after RQ-PO2 and is not linked to the RQ-PO2. 2. This training should come after a PO2 has worked as a MAR TECH PO2 on ship. It comes before the RQ-PO1. 3. To be posted as an electrical manager on ship, the RQ-PO1 is a required qualification as is the rank of PO1. 4. Many of the tasks listed within this training are for electrical managers versus electrical supervisors. 5. There is currently no course for electrical managers, however day one on the job as an electrical supervisor, the graduates will not be expected to do electrical manager tasks such as: a. Power isolation b. Post-incident reports c. Coordinating casualty power Email was sent with proposed COAs.
	PO 003 Submit Post Incident Report: this may not fit within this training. 1. This is generic and could be for both mechanical and electrical. 2. It is not at the PO2 level and is more geared towards PO1. After discussion with CPO2 Siska CCD Manager at NTDC(P), his recommendation was to remove this PO in entirety from the training program. This is still requiring approval from the TM/OM. (*see email above*). All tasks associated with PO 003 were changed to "No Train" in Annex B.
	Added the NQUAL ALUY and adjusted the job title to be "Electrical Supervisor (Elec Sup)" based on document from TM:
	Added and amended: standard statement for PO 001 Report on ship's PG&D system to include adding a. Advising command on ship's PG&D system health to include:

	(1) Equipment deficiencies; and (2) Risk mitigation COAs for CM and PM IAW MBS This is a No train task as it is achieved based on competencies from EO 001.01 and 001.02 as well as on the job. It is, however, part of the performance for an Elec Sup on the job.
	Changed “kill cards” in PO 001 standard and EO 001.01 to “Power isolation resources” which includes kill cards or power isolation books depending on the platform.
	Adjusted PO 002 standard: • Changed “Analyze power Isolation Conference” to “Brief DC Organization on Power isolation requirements”. This is because the Elec Sup is not required to attend or analyze the Power Isolation conference, but does provide information to the organization related to power isolation. This change also affects EO 002.02 which will now be “Brief DC Organization on Power Isolation Requirements.” • Removed “Identify Ship’s priority including float, move, fight”. Instead, added “Analyze power isolation requirements for compartments or zones affected by the event in relation to command priorities.” • Changed “Identify pipe” to “make power isolation pipe” as the Elec Sup is the one making the pipe. • Changed “Analyze reverse power isolation conference” to “Coordinate reverse power isolation conference” because the Elec Sup is actually doing the coordination and not just analyzing theoretically. This change will impact ~~EO 002.04~~ EO 002.02 Coordinate reverse power isolation conference.
	Removed “If abnormal…” section from EC 001.02. It will be taught, but it is unfair to have extra criteria for an abnormal route. Assessment should be IAW CFCD 132 routes.
	Adjusted EC 001.02 checklist to meet the standard for the EO.
	Combined EO 002.01 Manage power isolation with EO 002.02 Analyze power isolation conference and EO 002.03 Supervise power isolation. EO 002.01 and 002.02 were being assessed as one EC and Brief DC Organization on Power isolation requirements (changed in the PO standard statement) fits under managing power isolation so the LOs for briefing became part of managing. The difference between “supervising” and “managing” power isolation was vague and not well defined therefore it was deemed the “supervise” could fall under “manage”. There were no teaching points lost and standards and time were not affected by the change. Any slight duplication of training (ex. briefing of individuals) was eliminated. The PO is now separated into doing power isolation and then reversing power isolation. The PO standard was not adjusted as the job remained the same.
	Adjusted EC 002.01 checklist to meet the standard for the EO.
	EC 002.01 timing was adjusted based on additional criteria from added EOs.
	EC 002.02 timing was changed based on SME input.

	Learning objective language and assessment criteria were aligned throughout with the standard and assessment.
	All EC and PC checklists amended to be “SAT on all checklist items” for successful completion. This is because an Elec Sup is expected to do all the functions on the job independently and therefore only required items are on the checklists.
	Adjusted Chapter 1 to reflect updates to QSP based on the review to include: • Course duration • Resources (added all required resources and personnel requirements) • Special instructions (removed “high/med/low” priority for training and made it all high priority based on safety)
	Adjusted Annex A to reflect current QSP material, and ensured refs were included in proper lesson specifications
	Adjusted Annex D scalar to reflect new PO and EO format
	Updated Annex F acronyms and abbreviations to be more inclusive of terms used throughout the document
	Adjusted Annex G sequencing and timings for new course construct
	Added PCOT resources to Annex H
	Did not complete Annex I as the template is out of date
	Adjusted Annex K to reflect new course construct.
Chapter 1 Update 27 Oct 22	Pre-Requisite amendment to include “Minimum rank PO2” (matches USQ PO2 Mech). As per direction from TM. RE USQ-Po2 Elec Course Load reques

ANNEX K - *ASSESSMENT PLAN*

EC / PC	Scope Covered	Purpose	How	Pass / Fail Criteria	Impact
EC 001.01	EO 001.01	Formative assessment covering advising command on ship's power configuration	Practical scenario based assessment.	SAT on all checklist items	Remedial training, re-test with different scenario
EC 001.02	EO 001.02	Formative assessment to coordinate casualty power requirements	Practical scenario based assessment.	SAT on all checklist items	Remedial training, re-test with different scenario
PC 001	PO 001	Summative assessment to report on PG&D systems	Practical scenario based assessment.	SAT on all checklist items	Remedial Training / Re-test with different scenario / TRB after unsuccessful retest
EC 002.01	EO 002.01	Formative assessment to manage power isolation	Practical scenario based assessment.	SAT on all checklist items	Remedial training, re-test with different scenario
EC 002.02	EO 002.02	Formative assessment to coordinate reverse power isolation conference	Practical scenario based assessment. (Linked to scenario from EC 002.01)	SAT on all checklist items	Remedial training, re-test with different scenario
PC 002	EO 002.01-002.02	Summative assessment to advise command on ship's power isolation	Practical scenario based assessment.	SAT on all checklist items	Remedial Training / Re-test with different scenario / TRB after unsuccessful retest